ZOOLOGY OF FINANCIAL MARKETS

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- ✓ Is my own and personal opinion
- ✓ It is meant to be solely for educational purposes
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- ✓ Does not constitute any kind of investment advice

www.svsamiti.org

A bit about myself

UNIVERSIDAD NACIONAL AUTONOMA DE MEXICO
POR MI RAZA HABLARA EL ESPIRITU

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1. Zoology of financial markets

Financial markets as an ecosystem

- All markets have:
  - Infrastructure: **geography**
  - Regulations: **Nature's constraints**
  - Investment opportunities: **food**
  - Agents: **species**

- All financial agents are related in the market:
  - Cooperative way: **symbiosis**
  - Competitive way: **parasitism, food chain**

wildscenics.photoshelter.com

- Agents adapt and change strategies:
  - Increase profits and reduce losses: **mutation**

- Only the most profitable strategies survive: **natural selection**

Agents: investors (UK)

- **Small savers**:
  - They deposit their money in banks, earning an interest rate

bonnerandpartners.com

- **Small investors**:
  - People with less than 85k GBP
    - Protected by the Financial Services Compensation Scheme
  - Can only invest in standard instruments
    - Information disadvantage → more legal protection

- **High net worth investors** (according to the Financial Conduct Authority)
  - People with more than 250k GBP in assets or annual income of at least 100k GBP
  - Can invest in standard and non-standard financial instruments
    - Information advantage → less legal protection

- **Institutional investors:**
  - Entities that pool money from multiple individual clients
  - They invest to increase their assets under management (AUM)
  - Examples: Pension funds, insurance companies, asset managers, hedge funds, sovereign wealth funds

Agents: banks

- **Retail banks**:
  - Their profit is lend rate – deposit rate
    - They receive deposits from savers
    - They lend money to consumers
  - Risk management: creditworthiness of clients
  - They move capital to where it is needed
    - From savers to consumers/entrepreneurs

residential.jll.co.uk

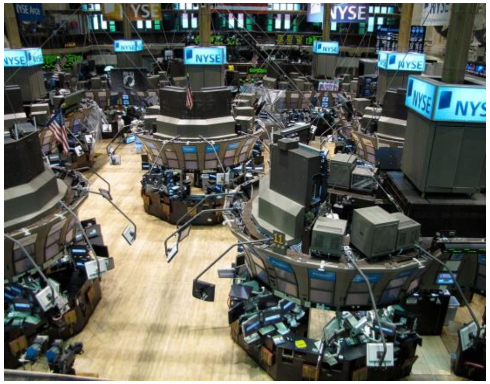
- **Investment banks**:
  - They sell financial instruments to investors: selling side
  - They create new financial products based on investors' demand
  - Complex products require strong quantitative analysts: **quants**
    - Mathematical and financial models
    - Risk management policies and forecasts
  - Profits made by volume traded + margin/prime over hedging price: flow trading

Agents: markets

- A market is a place where a buyer of an instrument meets a seller
  - It can be an exchange or over-the-counter

- A transaction in an **exchange**:
  - Done with listed, standard instruments
    - Shares, bonds, futures, options, ETFs
  - Buyers and sellers do not meet
    - They trade based on what is available in the market
  - Has a clearing house:
    - Daily margins and settlements
    - It minimises counterparty and credit risk

chapelleconsulting.com



- A transaction **OTC**:
  - Done with ad hoc, tailored financial instruments
    - Exotic options, structured products, CDOs
  - Buyers and sellers meet and agree to trade
  - Has no guarantee both sides will honour the agreement:
    - Counterparty and credit risk

www.entrepreneur.com

Agents: brokers

- They are **intermediaries** between investors and the markets
  - They buy/sell financial instruments on exchanges/OTC on behalf of clients

- They can also give financial and investment advice to clients
  - Suggest products based on clients' risk appetite
  - Provide research on stocks and market trends

stockbrokerpro.com

- They charge **commissions** for their services:
  - Account management: annual fee
  - Per deal: unitary price
  - Per volume: fraction of total notional traded

- Their risk is more reputational than monetary
  - The profit and loss (PNL) on a trade sits totally on the client's side
  - Brokers live by the commissions: no volume → no business
  - Investors choose brokers for their trustworthiness and services

Agents: market makers

- They are **liquidity providers**
  - They offer buy and sell prices to traders and investors
- They make firm buy and sell quotes:
  - They show a price and a volume on both buy and sell
  - They are bound to trade at those conditions

youtube.com – investor trading academy

- Their profit is made via the **spread**:
  - The spread is sell price – buy price
  - If there is a buy (sell) followed by a sell (buy), they earn the spread
  - If there is enough flow on both buys and sells, they are profitable

quantinsti.com

- Their risk profile has two components
  - **Adverse selection**: they are always on the wrong side of a trend
  - **Inventory risk**: they cannot (fully) control the arrival of trade orders
- Side note: the spread is a proxy of liquidity
  - More liquid names → more market makers competing → tighter spreads

Agents: asset managers

- They are professionals of portfolio management

- They invest on financial instruments on behalf of their clients
  - High net worth individuals, institutional investors
  - Small investors via pooling and investment vehicles

keywordsuggest.org/gallery

- They charge an annual fee based on the AUM
  - Between 0.5% and 2%

- They invest based on a given mandate
  - Index trackers, long only, long/short, leveraged, etc
  - Equities, fixed income, cash, property, etc
  - Allocation based on risk appetite

- **Benchmark**: target return for the fund (e.g. the return of the S&P 500)
  - As long as the manager beats the benchmark, they are doing well
  - Tracker funds try to replicate the benchmark's performance, not beat it

Agents: hedge funds

- They are asset management firms with alternative investment strategies

- More liberty in their investment toolkit
  - Leveraged products
  - Long/short
  - Derivatives
  - High turnover

- They charge 2 kinds of fees:
  - Like asset managers, an annual fee: 2%
  - Performance fee:
    - The fund keeps 20% of all profits
    - But only if the fund is profitable

- They aim for absolute return
  - Aim to deliver positive returns in all market conditions
  - Not a relative return vs benchmark, as asset managers

cnbc.com

Agents: prime brokers

- They are the credit lines of the investment management firms
  - This service is mainly provided by investment banks

psihologiya.top

- Some of the services they offer:
  - Borrow money, so investors can trade
  - Borrow stock, so investors can short sell
  - Advisory and support:
    - Legal, accounting, commercial

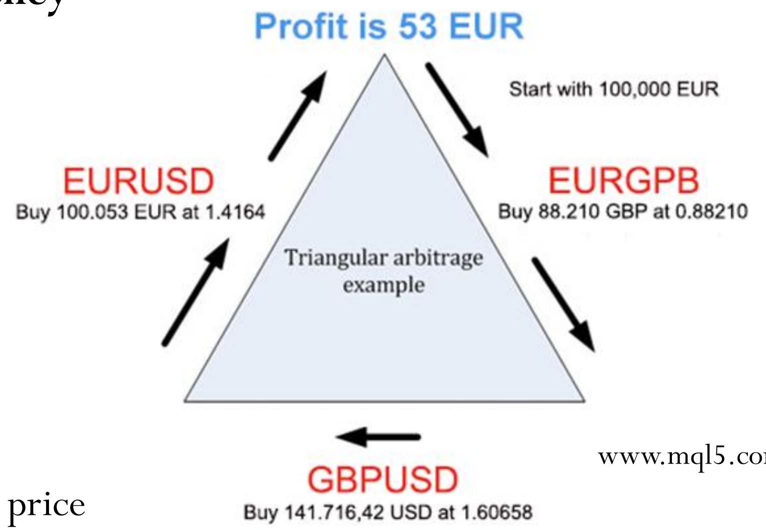
- Access to a wide range of products:
  - All products and services in the prime broker's firm
    - Execution, research, structured products, swaps, funds, ETFs, etc

Agents: arbitrageurs

- An arbitrage is to **take advantage of a market inefficiency**

- An arbitrageur
  - Generally a short-term speculator
  - But they make markets more efficient

- Examples of arbitrage:
  - **Prices** or mispricing arbitrage:
    - Buy and sell instruments far from their fundamental price
  - **Venues** or space arbitrage:
    - Buy and sell the same product in two different markets, cashing the difference
  - **Time** value or term-structure arbitrage:
    - Interest rates and volatility; calendar strategies with options
  - **Statistical** arbitrage or stat arb:
    - Spot and exploit trading patterns that are statistically robust



www.mql5.com

Agents: prop traders

- They speculate in capital markets with **their own money**
  - Unlike funds, they have no clients

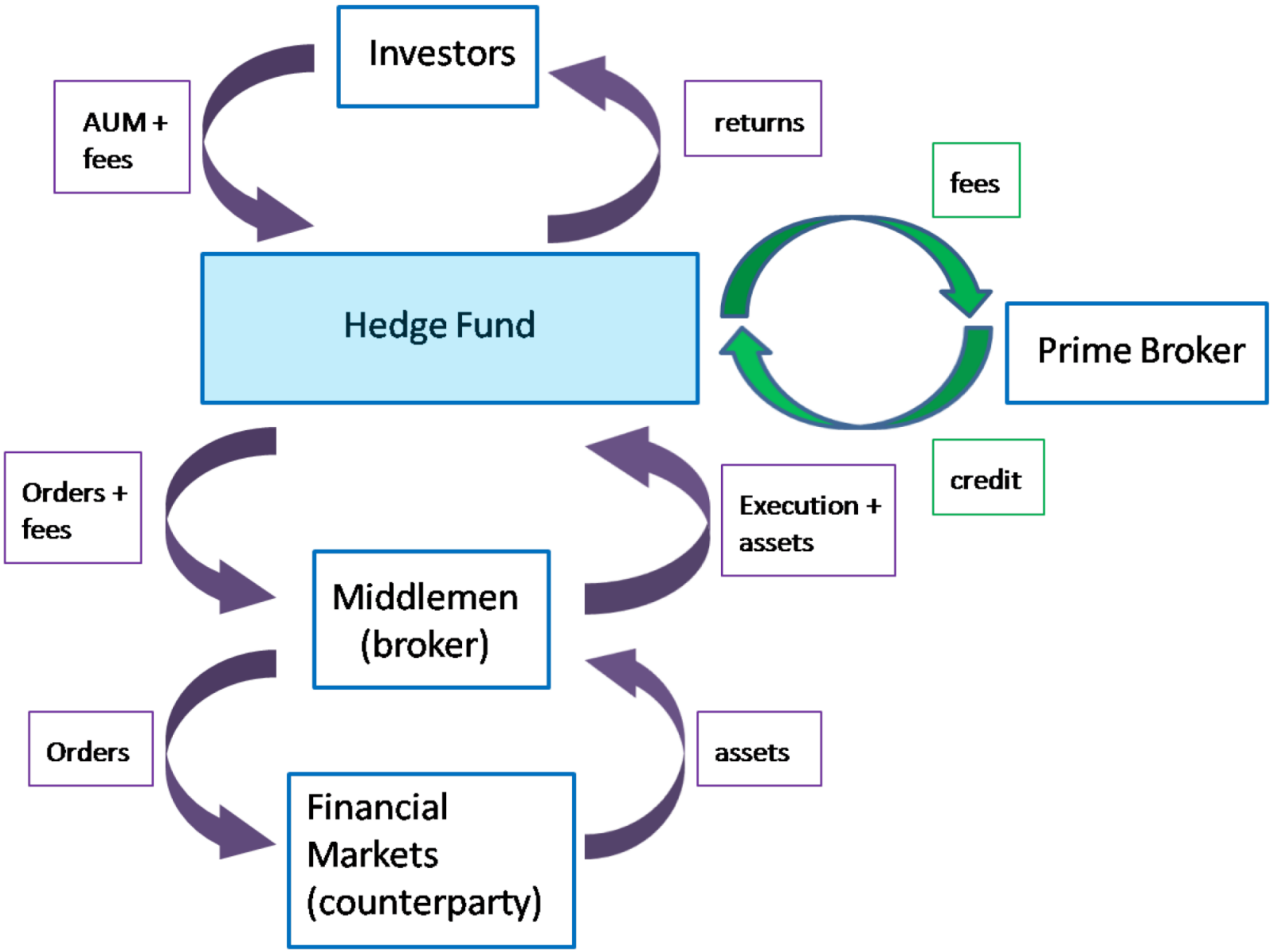
- They have the **highest flexibility** on investment strategies:
  - All asset classes: equities, fixed income, forex, credit, commodities
  - All markets: US, Europe, APAC, Emerging, etc
  - All derivatives: vanilla, exotic, structured, etc

www.sanglucci.com

- Particular case of prop trading: **High Frequency Trading**
  - Investing in technology and speed for a competitive advantage
  - Latency and reaction time measured in microseconds
  - Examples of HFTs: arbitrageurs and market makers

- Before the Dodd-Frank reform and the Volcker Rule:
  - Banks used to have their own prop trading desks
- Currently, prop trading in banks is almost banned
  - But market makers and risk traders are exonerated of the Volker rule

The financial ecosystem in one picture



2. Investment funds and CAPM

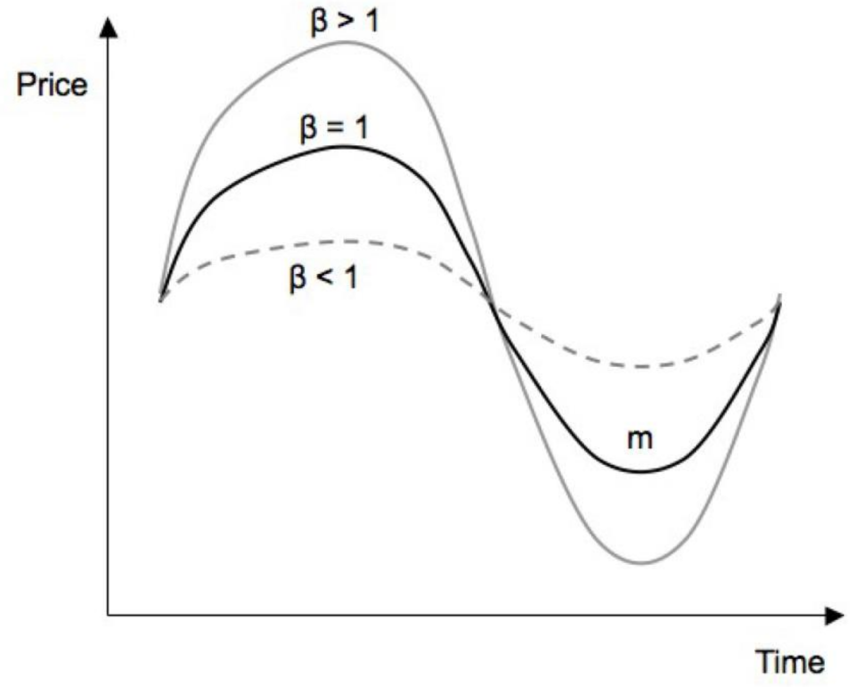
Capital Asset Pricing Model (CAPM)

- The CAPM relates the return of an asset to the return of the market

- Define
  - r_a the return of asset a
  - r_M the return of the market M
  - r_F the return of the risk-free rate F

- Then
  - $\mathrm{E}[r_a] - r_F = \beta(\mathrm{E}[r_M] - r_F)$

- Where
  - $\beta = \frac{Cov(r_a, r_M)}{Var(r_M)}$



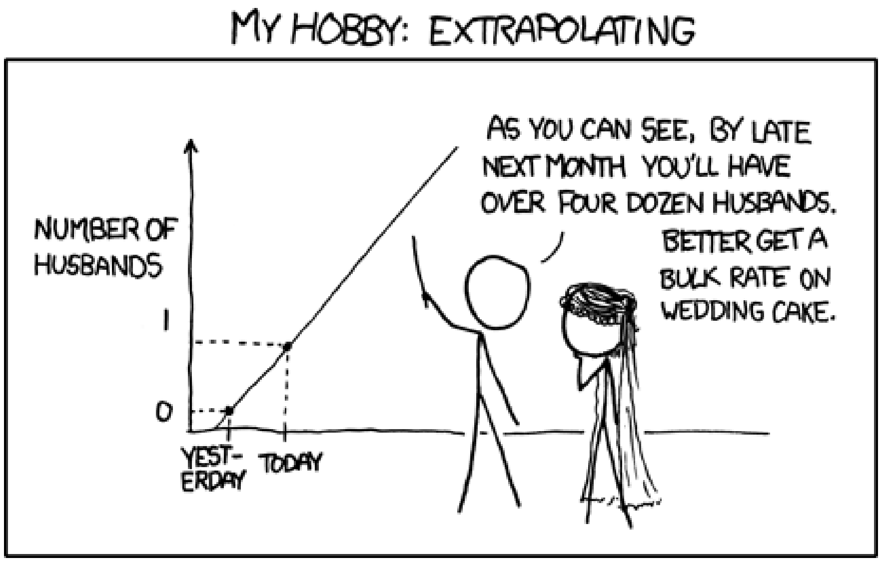
CAPM, alpha and beta

- In a sentence, the CAPM is a linear regression of r_a w.r.t. r_M

- A classical linear regression with intercept is
  - $r_a = \alpha + \beta r_M + \varepsilon$

- The error ε is called the **idiosyncratic risk**
  - It can be eliminated via diversification
  - ε is orthogonal to r_M and $E[\varepsilon] = 0$

- β is called the **systematic risk**
  - Exposure of asset a to the market
  - Cannot be diversified away



imgs.xkcd.com

- α is called the **absolute return**
  - α is orthogonal to r_M and ε
  - It is the fraction of the total return r_a that cannot be explained by market fluctuations

Classifying funds with CAPM

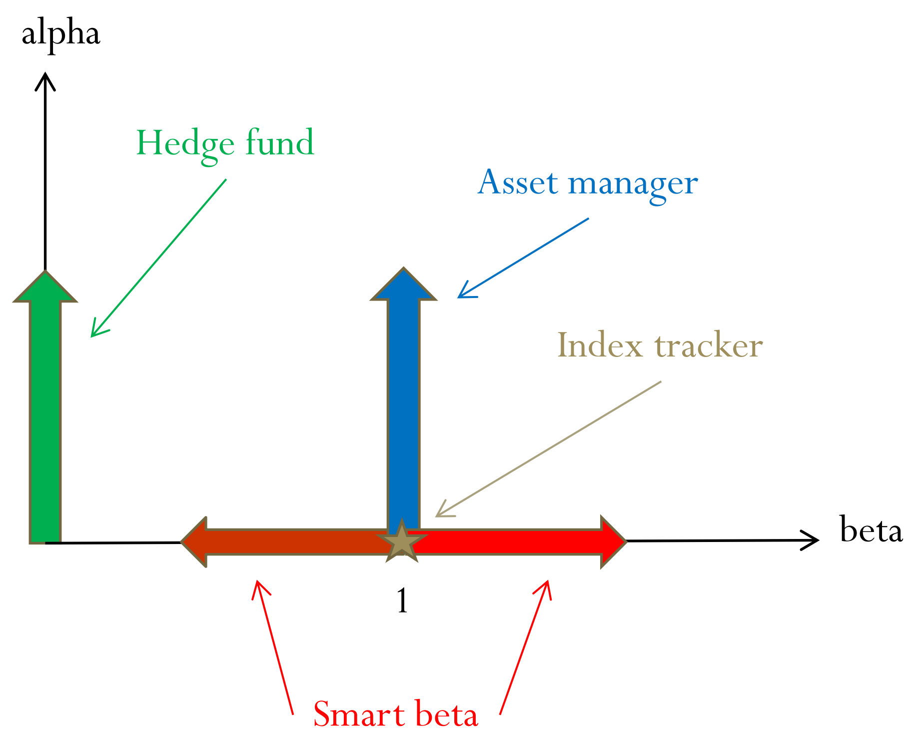
- **Index tracker**: Replicate the performance of a benchmark e.g. an index such as S&P 500
  - $\beta = 1$
  - $\alpha = 0$

- **Asset manager:** Outperform the market by adding some uncorrelated extra return
  - $\beta = 1$
  - $\alpha > 0$

- **Smart beta:** Outperform the market by slightly modifying the market weights of an index
  - $\beta > 1$ when the market is up
  - $\beta < 1$ when the market is down
  - $\alpha = 0$

- **Hedge fund**: Deliver absolute returns not correlated with the market
  - $\beta = 0$
  - $\alpha > 0$

Diagram of funds

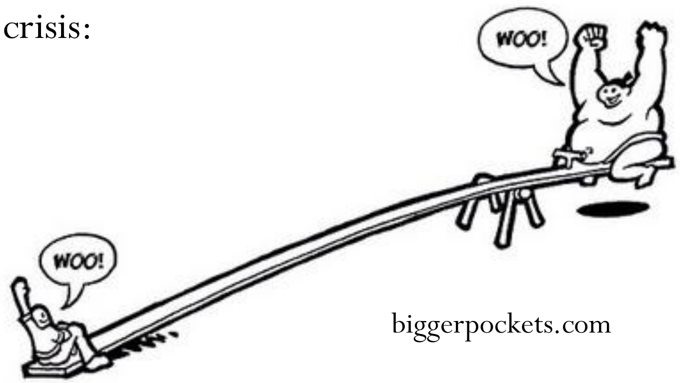


3. Some investment strategies

Leverage

- The leverage is the ratio of the total value of the positions over the total assets under management (AUM)
  - $Leverage = \frac{Positions}{AUM}$

- According to the OECD (OCDE in Spanish), during the 2007 crisis:
  - The investment banks had an average leverage of 30
- The Basel III rules require at least 10.5% of liquid assets
  - Maximum leverage of 9.5
- Historically, hedge funds have an average leverage of 5



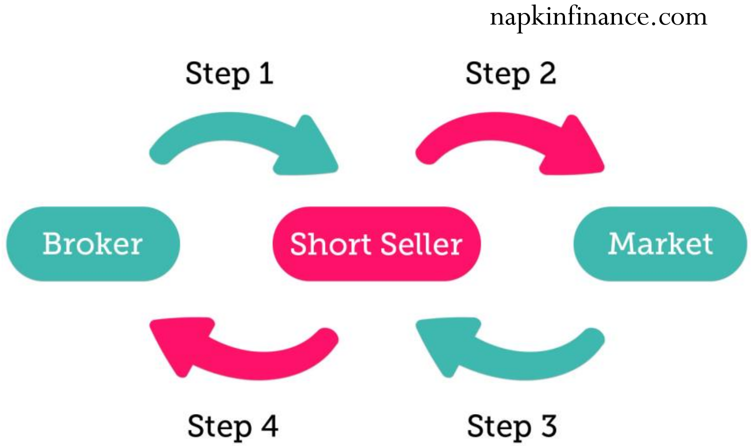
biggerpockets.com

- Leverage is a double-edged sword
  - It amplifies the profits
  - But also amplifies the losses

- Some Forex websites offer leverages up to 400%
  - A gain of 0.10% in the trade → a profit of $0.1\% * 400 = 40\%$ in the investment
  - A loss of 0.25% in the trade → a loss of $0.25\% * 400 = 100\%$ in the investment

Short selling

- A short sell is a trading strategy
  - It is a bet that a security (usually a stock) will **fall in price**
  - We sell an asset now with the idea of buying it later at a cheaper price, making a profit



- It is often labelled as "selling something you do not own"
- But it is more than that:
  - Step 1: Borrow a stock from your prime broker
  - Step 2: Sell the borrowed stock on the market
  - Step 3: Buy back the stock on the market
  - Step 4: Return the borrowed stock to the prime broker
    - Plus an **interest rate** for the loan

- Short selling is an asymmetric bet due to the loan rate:
  - Suppose the loan rate is 0.10%
  - If the price goes down by 1% → the net profit is 0.9%
  - If the price goes up by 1% → the net loss is 1.1%

Derivatives

- A derivative is a financial instrument whose value depends on another instrument
  - The **underlying**

quora.com

- Examples:
  - Futures, options, ETFs, swaps, CDSs, CDOs, etc

- Properties:
  - They allow investors to have specific return profiles (payoff)
  - Their price depends on several variables and their sensitivities (derivatives or **greeks**):
    - Delta: price of the underlying
    - Vega: volatility of the underling
    - Rho: interest rates
    - Theta: time
    - Gamma: changes in the delta
  - Some derivatives have an intrinsic leverage effect

Example of leverage in options

- Spot price: $S(t) = 100$ GBP
- Strike Price: $K = 100$ GBP (at-the-money option)

- Assume $S(t)$ follows a **binomial tree** of period 1:
  - Price can go up to 103 with probability p
  - Price can go down to 98 with probability $1 - p$

- **Risk-neutral probability**: p such that $S(t)$ is a martingale
  - $E[S(t+1)|F_t] = S(t) = 100$
- Binomial tree:
  - $E[S(t+1)|F_t] = p * 103 + (1-p) * 98$
- Therefore $100 = p * 103 + (1-p) * 98 \rightarrow p = 2/5$

- **Call option price**:
  - $E[(S(t+1) - K)_+|F_t] = \frac{2}{5} * (103 - 100)_+ + \frac{3}{5} * (98 - 100)_+ = 1.20$ GBP
  - Price up 3% → option at $t + 1$ is worth 3 GBP → return of $\frac{3-1.2}{1.2} = +150\%$
  - Price down 2% → option at $t + 1$ is worthless → return of $\frac{0-1.2}{1.2} = -100\%$

4. Conclusions

Summary and conclusions

- Summary
  - We described the different "species" in the "zoology" of the financial markets
  - We showed how the agents interact between themselves
  - We revisited the CAPM and showed how it can be used to classify investment funds
  - We explained 3 investment strategies: leverage, short selling and derivatives

- Conclusions
  - There are several different agents in the financial markets
  - The beauty of Finance is that there is a niche for everyone
    - Trading styles
    - Investment strategies
    - Asset classes
  - But not all strategies, styles and asset classes are adequate for every one
    - Choose the best combination based on your personal risk profile

Thank you for your attention

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